Enzo Guerrero-Araya, Ph.D.

Phylodynamics – Epidemiology – Computational Biology – Data Science

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Experience

Nov 2023 – Present — **Research Fellow, University of Nottingham Ningbo China.** Conduct research in genomic epidemiology with emphasis on mobile genetic elements and microbial populations. Use computational genomics and bioinformatics workflows to analyze, interpret, and communicate genomic data relevant to public health.

Feb 2021 – Feb 2023 — **Data Scientist, Fundacion Ciencia & Vida.** Prepared, mined, visualized, and analyzed COVID-19 pandemic data, contributing data-driven evidence on viral spread and impact.

Mar 2017 – Jul 2017 — **Teaching Experience, Andres Bello University.** Professor for the Laboratory of Genetic Engineering in Microorganisms (BIT200), guiding students in genetic engineering techniques and their application in microorganisms.

Education

2018 – 2023 — **Doctor of Philosophy, Andres Bello University.** During my Ph.D. project I studied the evolution and genomic epidemiology of *C. difficile* in Latin America. International collaborations in evolution, bioinformatics, and data analysis led to scientific publications.

2012 – 2016 — **Bachelor of Biotechnology, Andres Bello University.** Training in microbiology and genetics, with specialization in bacterial genome assembly, genomic data analysis, and bioinformatics.

Publications

Submitted

1. **E. Guerrero-Araya**, F. Cid-Rojas, M. Munoz, C. Rodriguez, and D. Paredes-Sabja, "Identification of novel cryptic and classical clades in *C. difficile*," 2026, Submitted. Enzo Guerrero-Araya led the analysis of 25,145 *C. difficile* genomes plus 21 new isolates, integrating MLST, ANI, and recombination-corrected core-genome phylogenies to identify C6/C7, C-VI/C-VII, and the C-IIIa/C-IIIb split.

Published

1. M. Baker, A. Maciel-Guerra, R. Wang, C. Luo, Y. Xu, **E. Guerrero-Araya**, W. Meng, G. Wu, K. Pinpimai, P. A. Oyom, N. Senin, and T. Dottorini, "Genomic and socioeconomic drivers of antimicrobial resistance forecast to 2050," *Cell Genomics*, vol. 6, p. 101 273, Jun. 2026, Enzo Guerrero-Araya led acquisition and analysis of bacterial genomes, identified ARGs, and annotated their genomic context. DOI: 10.1016/j.xgen.2026.101273
2. F. Cid-Rojas, **E. Guerrero-Araya**, C. Brito-Silva, M. Pizarro-Guajardo, C. Rodriguez-Sanchez, and D. Paredes-Sabja, "Phylogenomic analysis of the collagen-like BclA proteins in *C. difficile*," *Applied and Environmental Microbiology*, 2025, Enzo Guerrero-Araya led phylogenomic analysis across 25,000 *C. difficile* genomes and contributed to AlphaFold3 structural modeling.
3. **E. Guerrero-Araya**, C. Ravello, M. Rosemblatt, and T. Perez-Acle, "Socioeconomic status correlates with covid-19 vaccination coverage among primary and secondary students in the most populated city of chile," *Scientific Reports*, 2025, Enzo Guerrero-Araya led the analysis linking socioeconomic status with COVID-19 vaccination coverage in over 1,600 schools in Santiago, Chile.
4. A. Camargo, **E. Guerrero-Araya**, S. Castaneda, L. Vega, M. X. Cardenas-Alvarez, C. Rodriguez, D. Paredes-Sabja, J. D. Ramirez, and M. Munoz, "Intra-species diversity of *C. perfringens*: A diverse genetic repertoire reveals its pathogenic potential," *Frontiers in Microbiology*, 2022, Enzo Guerrero-Araya conducted in silico MLST assignment and phylogenetic reconstruction.
5. L. H. Patino, S. Castaneda, M. Munoz, N. Ballesteros, A. L. Ramirez, N. Luna, **E. Guerrero-Araya**, J. Perez, C. A. Correa-Cardenas, M. C. Duque, C. Mendez, C. Oliveros, M. V. Shaban, A. E. Paniz-Mondolfi, and J. D. Ramirez, "Epidemiological dynamics of SARS-CoV-2 variants during social protests in cali, colombia," *Frontiers in Medicine*, 2022, Enzo Guerrero-Araya calculated the effective reproductive number of SARS-CoV-2 over time in Colombia.
6. S. Castaneda, L. H. Patino, M. Munoz, N. Ballesteros, **E. Guerrero-Araya**, D. Paredes-Sabja, C. Florez, S. Gomez, C. Ramirez-Santana, G. Salguero, J. E. Gallo, A. E. Paniz-Mondolfi, and J. D. Ramirez, "Evolution and epidemic spread of SARS-CoV-2 in colombia: A year into the pandemic," *Vaccines*, 2021, Enzo Guerrero-Araya calculated the effective reproductive number of SARS-CoV-2 over time in Colombia.

7. **E. Guerrero-Araya**, M. Munoz, C. Rodriguez, and D. Paredes-Sabja, "FastMLST: A multi-core tool for multilocus sequence typing of draft genome assemblies," *Bioinformatics and Biology Insights*, 2021, Co-corresponding author. Enzo Guerrero-Araya designed, developed, tested, implemented FastMLST, and drafted the manuscript.
8. D. R. Knight, K. Imwattana, B. Kullin, **E. Guerrero-Araya**, D. Paredes-Sabja, X. Didelot, K. E. Dingle, D. W. Eyre, C. Rodriguez, and T. V. Riley, "Major genetic discontinuity and novel toxigenic species in *C. difficile* taxonomy," *eLife*, 2021, Enzo Guerrero-Araya performed Bayesian evolutionary analysis, molecular dating, toxin detection, and genomic-context analysis.
9. **E. Guerrero-Araya**, C. Meneses, E. Castro-Nallar, A. M. Guzman D., M. Alvarez-Lobos, C. Quesada-Gomez, D. Paredes-Sabja, and C. Rodriguez, "Origin, genomic diversity and microevolution of the *C. difficile* B1/NAP1/RT027/ST01 strain in costa rica, chile, honduras and mexico," *Microbial Genomics*, 2020, Enzo Guerrero-Araya performed assembly, annotation, evolutionary analysis, prepared figures and tables, and drafted the manuscript.
10. M. Munoz, **E. Guerrero-Araya**, C. Cortes-Tapia, A. Plaza-Garrido, T. D. Lawley, and D. Paredes-Sabja, "Comprehensive genome analyses of *S. intestinalis*, a potential biomarker of homeostasis gut recovery," *Microbial Genomics*, 2020, Enzo Guerrero-Araya performed Bayesian evolutionary analysis and protein annotation using Clusters of Orthologous Groups.
11. A. Romero-Rodriguez, S. Troncoso-Cotal, **E. Guerrero-Araya**, and D. Paredes-Sabja, "The *C. difficile* cysteine-rich exosporium morphogenetic protein, CdeC, exhibits self-assembly properties that lead to organized inclusion bodies in *E. coli*," *mSphere*, 2020, Enzo Guerrero-Araya performed evolutionary and bioinformatics analyses and designed figures.
12. P. Calderon-Romero, P. Castro-Cordova, R. Reyes-Ramirez, M. Milano-Cespedes, **E. Guerrero-Araya**, M. Pizarro-Guajardo, V. Olguin-Araneda, F. Gil, and D. Paredes-Sabja, "*C. difficile* exosporium cysteine-rich proteins are essential for the morphogenesis of the exosporium layer, spore resistance, and affect *C. difficile* pathogenesis," *PLoS Pathogens*, 2018, Enzo Guerrero-Araya performed evolutionary and bioinformatics analyses and designed figures.
13. **E. Guerrero-Araya**, A. Plaza-Garrido, F. Diaz-Yanez, M. Pizarro-Guajardo, S. L. Valenzuela, C. Meneses, F. Gil, E. Castro-Nallar, and D. Paredes-Sabja, "Genome sequence of *C. paraputrificum* 373-a1 isolated in chile from a patient infected with *C. difficile*," *Genome Announcements*, 2016, Enzo Guerrero-Araya performed assembly, identification, annotation, prepared figures and tables, and drafted the manuscript.

Skills

Research	Academic writing; bioinformatics; genomic sequence analysis; phylogenomics; microbiology; self-learning.
Data	Data science and analysis; data visualization; reproducible analysis workflows.
Programming	Python; R; Rust; Bash; LaTeX.
Systems	Cloud computing; high-performance computing; UNIX systems administration.
Languages	Spanish and English.

Posters

2019 — **Diversity, origin, and microevolution of *C. difficile* B1/NAP1/RT027/ST01 strains from Latin America. Enzo Guerrero-Araya**, Claudio Meneses, Eduardo Castro-Nallar, Ana M. Guzman D., Manuel Alvarez-Lobos, Carlos Quesada-Gomez, Daniel Paredes-Sabja, Cesar Rodriguez. CLOSTPATH 11, Stadsgehoorzaal, Leiden, The Netherlands.

2015 — **Identification of the pathway of *C. difficile* spore-entry into intestinal epithelial cells.** Pablo Castro-Cordova, **Enzo Guerrero-Araya**, Glenda Cofre-Araneda, Daniel Paredes-Sabja. XXXVIII Annual Meeting, Sociedad de Bioquimica y Biologia Molecular de Chile, Puerto Varas, Chile.